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import time

class Environment:
    def __init__(self):
        # Take user input for the environment
        self.locations = {}

        for location in ['A', 'B']:
            while True:
                status = input(
                    f"Enter status of Location {location} (Clean/Dirty): "
                ).strip().capitalize()

                if status in ['Clean', 'Dirty']:
                    self.locations[location] = status
                    break
                else:
                    print("Invalid input! Enter Clean or Dirty.")

        # Take initial vacuum cleaner position
        while True:
            position = input(
                "Enter initial vacuum cleaner position (A/B): "
            ).strip().upper()

            if position in ['A', 'B']:
                self.agent_location = position
                break
            else:
                print("Invalid input! Enter A or B.")

        self.total_cost = 0

    def get_percept(self):
        return (
            self.agent_location,
            self.locations[self.agent_location]
        )

    def execute_action(self, action):
        print(
            f"Agent is at Location {self.agent_location} | "
            f"Status: {self.locations[self.agent_location]} | "
            f"Action taken: {action}"
        )

        if action == 'Suck':
            self.locations[self.agent_location] = 'Clean'
            self.total_cost += 1

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        self.total_cost += 1

    elif action == 'Right':
        self.agent_location = 'B'
        self.total_cost += 1

    elif action == 'Left':
        self.agent_location = 'A'
        self.total_cost += 1

    elif action == 'NoOp':
        pass

def are_all_locations_clean(self):
    return (
        self.locations['A'] == 'Clean'
        and self.locations['B'] == 'Clean'
    )

class SimpleReflexVacuumAgent:
    def reflex_agent_program(self, percept):
        location, status = percept

        if status == 'Dirty':
            return 'Suck'

        elif location == 'A':
            return 'Right'

        elif location == 'B':
            return 'Left'

        return 'NoOp'

# --- Simulation Runner ---

if __name__ == "__main__":

    print("Name: HRITHIKA RAJESH")
    print("USN: 1BM24CS120")
    print("-" * 40)

    env = Environment()
    agent = SimpleReflexVacuumAgent()

    print("\nInitial Environment State:", env.locations)
    print(f"Vacuum starting at location: {env.agent_location}")
    print("-" * 40)

    steps = 0

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max_steps = 10

while steps < max_steps:

    # Stop when both locations are clean
    if env.are_all_locations_clean():
        print("All locations are clean! Stopping agent.")
        break

    current_percept = env.get_percept()

    chosen_action = agent.reflex_agent_program(
        current_percept
    )

    env.execute_action(chosen_action)

    steps += 1
    time.sleep(0.5)

print("-" * 40)
print("Final Environment State:", env.locations)
print(f"Final Vacuum Position: {env.agent_location}")

print(
    f"FINAL PERFORMANCE SCORE -> "
    f"Total Cost Incurred: {env.total_cost} points"
)

```

Name: HRITHIKA RAJESH

USN: 1BM24CS120

Enter status of Location A (Clean/Dirty): Dirty
Enter status of Location B (Clean/Dirty): Dirty
Enter initial vacuum cleaner position (A/B): A

Initial Environment State: {'A': 'Dirty', 'B': 'Dirty'}
Vacuum starting at location: A

Agent is at Location A | Status: Dirty | Action taken: Suck
Agent is at Location A | Status: Clean | Action taken: Right
Agent is at Location B | Status: Dirty | Action taken: Suck
All locations are clean! Stopping agent.

Final Environment State: {'A': 'Clean', 'B': 'Clean'}
Final Vacuum Position: B
FINAL PERFORMANCE SCORE -> Total Cost Incurred: 3 points

